

A Likelihood-Free Fiducial Framework for Estimating Equation Models

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Abstract

Estimating equations provide a flexible framework for statistical inference in settings where a full likelihood specification is unavailable or undesirable. However, existing uncertainty quantification for estimating-equation estimators typically relies on first-order asymptotic approximations or computationally intensive resampling procedures. This paper develops a likelihood-free fiducial framework for constructing confidence distributions in general estimating-equation models. Rather than requiring an explicit structural equation for the observed data, the proposed method constructs uncertainty directly from the empirical estimating equations. A Gaussian copula representation is used to generate stochastic perturbations that preserve the estimated dependence structure of the empirical moments, which are subsequently inverted to obtain fiducial draws of the parameter. Unlike procedures that directly simulate the asymptotic distribution of an estimating-equation estimator, CGFI treats the perturbed estimating equations as a stochastic inverse problem and constructs uncertainty directly on the parameter space. The resulting Copula Generalized Fiducial Inference (CGFI) procedure provides a general confidence distribution for estimating-equation parameters without requiring either a likelihood function or prior specification. Under standard regularity conditions, the proposed method is shown to be consistent and to satisfy a Bernstein–von Mises-type property. The resulting first-order stochastic expansion provides a foundation for future higher-order refinements. Simulation studies demonstrate that CGFI achieves coverage probabilities comparable to bootstrap procedures while substantially reducing computational cost in many-moment estimating-equation settings. The proposed framework extends generalized fiducial inference to a broad class of estimating-equation models where likelihood-based inference is unavailable or undesirable and provides a scalable approach to uncertainty quantification for modern estimating-equation methods.

Keywords: Confidence distributions; Estimating equations; Generalized fiducial inference; Gaussian copula; Generalized method of moments; Likelihood-free inference.

1 Introduction

Statistical inference aims to quantify uncertainty about unknown parameters based on observed data. Classical frequentist methods achieve this through sampling distributions of estimators and confidence procedures, while Bayesian methods represent uncertainty through posterior distributions obtained by combining likelihood functions with prior distributions. Both frameworks typically rely on an explicit probabilistic specification of the data-generating process.

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Fiducial inference, introduced by Fisher (1930, 1935), provides an alternative approach by constructing a distribution for an unknown parameter without introducing prior information. The fundamental idea is to express the data-generating mechanism through a structural equation,

$$X = G(U, \theta),$$

where U represents auxiliary randomness with a known distribution. After observing X , the relationship is inverted to obtain a distribution for θ induced by repeated realizations of U . In simple parametric settings, this construction reproduces classical confidence procedures. For example, under a normal location model with known variance,¹

$$\bar{X} = \mu + \frac{\sigma}{\sqrt{n}}Z,$$

where $Z \sim N(0, 1)$, inversion gives

$$\mu = \bar{X} - \frac{\sigma}{\sqrt{n}}Z,$$

and the resulting fiducial distribution has quantiles corresponding to the usual normal confidence interval.

Despite its intuitive appeal, classical fiducial inference is difficult to extend beyond simple models because many statistical procedures do not admit a unique or natural structural representation. Generalized fiducial inference (GFI), developed by Hannig (2009) and further studied by Hannig et al. (2016), addresses these limitations by providing a broader framework based on approximate inversion of stochastic data-generating equations. Generalized fiducial methods have since been applied to a wide variety of parametric and semiparametric models while retaining a frequentist interpretation through confidence distributions.

However, many modern statistical procedures are not defined through explicit data-generating equations. Instead, parameters are characterized through estimating equations of the form

$$E\{g(X, \theta)\} = 0.$$

This framework includes generalized method of moments (Hansen, 1982), M-estimation (Huber, 1964), generalized estimating equations (Liang and Zeger, 1986), and a broad class of semiparametric estimators (Newey and McFadden, 1994). Estimating equations are particularly attractive because they require minimal assumptions on the underlying distribution of the data while allowing flexible modeling of complex relationships. Nevertheless, the absence of an explicit structural equation creates a fundamental difficulty for generalized fiducial inference: the usual inversion step is no longer directly available. Consequently, there is currently no broadly applicable fiducial framework specifically designed for constructing confidence distributions when parameters are defined only through general estimating equations.

This paper develops Copula Generalized Fiducial Inference (CGFI), a likelihood-free confidence distribution framework for parameters defined through estimating equations. The key idea is to

¹The assumption that the variance is known is made only for expositional simplicity. When the variance is unknown, the fiducial construction proceeds by replacing the normal pivot with the Student's t pivot,

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1},$$

where S is the sample standard deviation. Inverting this relationship yields the classical t confidence interval, demonstrating that the fiducial argument extends naturally to the unknown-variance setting (Fisher, 1935).

replace the unavailable structural equation for the observations with a stochastic representation of the empirical estimating equations themselves. Specifically, we treat

$$\bar{g}_n(\theta) = \frac{1}{n} \sum_{i=1}^n g(X_i, \theta)$$

as the fundamental inferential object and construct uncertainty by perturbing the empirical estimating equations according to their estimated sampling distribution. Under standard regularity conditions, the empirical estimating equations satisfy a multivariate central limit theorem, providing the stochastic mechanism required for generalized fiducial inversion.

A central feature of CGFI is that the stochastic perturbation is not used merely to approximate the sampling distribution of an estimator. Instead, the perturbed estimating equations define a collection of auxiliary stochastic systems that are inverted to obtain a distribution directly on the parameter space. The resulting fiducial distribution is therefore generated through stochastic inversion rather than through repeated approximation of the sampling distribution of an estimator.

The resulting procedure differs from existing likelihood-free and resampling approaches. Unlike approximate Bayesian computation and synthetic likelihood methods, CGFI does not require an observation-level simulator or a generative model for the data. Instead, it operates directly on estimating equations that characterize the parameter of interest. Unlike generalized Bayesian methods, it does not introduce prior distributions or loss-based updating rules. Unlike bootstrap procedures, which approximate the sampling distribution of an estimator through repeated resampling, CGFI constructs a confidence distribution by directly inverting a stochastic representation of the estimating equations.

The proposed framework is related to several existing approaches for inference in models where the likelihood function is unavailable, including approximate Bayesian computation, synthetic likelihood methods, generalized Bayesian inference, empirical likelihood, and resampling procedures. These methods share the goal of constructing uncertainty quantification procedures without requiring a fully specified likelihood, but they differ substantially in their inferential objectives and construction.

Approximate Bayesian computation (ABC) avoids likelihood evaluation by simulating data from a specified generative model and accepting parameter values that produce simulated summaries close to the observed summaries (Beaumont et al., 2002; Marin et al., 2012). Synthetic likelihood methods similarly rely on simulation but approximate the likelihood of selected summary statistics through a parametric distribution, typically Gaussian (Wood, 2010). In contrast, CGFI does not require a generative model for the observations. Instead, it operates directly on estimating equations that characterize the parameter of interest.

CGFI is also related to generalized Bayesian approaches that replace the likelihood with a loss function or moment-based criterion (Bissiri et al., 2016; Chernozhukov and Hong, 2003). These approaches provide useful alternatives when likelihood specifications are unavailable but retain a Bayesian interpretation through prior distributions or loss-based updating. CGFI differs by constructing a confidence distribution through generalized fiducial inversion without requiring prior information.

Empirical likelihood and related methods also use estimating equations as the basis for inference (Owen, 2001). However, empirical likelihood constructs uncertainty by assigning probabilities to observations subject to moment restrictions, whereas CGFI constructs uncertainty by perturbing the empirical estimating equations according to their sampling distribution and inverting the resulting stochastic system.

Finally, CGFI is closely connected to multiplier bootstrap methods, which introduce randomness through empirical estimating equations (Præstgaard and Wellner, 1993; van der Vaart and

Wellner, 1996). Although CGFI and multiplier bootstrap methods both introduce stochastic perturbations of empirical estimating equations, their inferential objectives are fundamentally different. Multiplier bootstrap methods use the perturbed estimating equations as an intermediate device for approximating the repeated sampling distribution of an estimator. In contrast, CGFI treats each perturbed estimating equation as a generalized pivotal relationship and performs stochastic inversion to obtain a distribution for the unknown parameter itself. Thus, the randomness in CGFI is not introduced to approximate an estimator distribution after the fact, but instead defines the confidence distribution through the inversion mechanism.

These distinctions are summarized in Table 1. Unlike existing likelihood-free and resampling approaches, CGFI does not require a likelihood function, prior distribution, or fully specified likelihood-based model, while still producing a confidence distribution through stochastic inversion of estimating equations.

Table 1: Comparison of CGFI with Related Likelihood-Free and Moment-Based Methods

Method	Requires Likelihood or Prior	Uses Estimating Equations	Produces Confidence Distribution
ABC	No likelihood, requires simulator	Indirectly	No
Synthetic likelihood	No likelihood, requires simulator	Through summaries	No
Generalized Bayes	Prior required	Often	No
Empirical likelihood	No	Yes	No
Bootstrap	No	Yes	No
Multiplier bootstrap	No	Yes	No
CGFI	No	Yes	Yes

The proposed framework provides a general approach for likelihood-free uncertainty quantification in estimating-equation models. The objective is not to replace existing asymptotic approximations or resampling procedures, but rather to provide a unified confidence distribution framework in settings where the parameter is identified through moment restrictions and an explicit likelihood or structural model is unavailable.

The contribution of this paper is fourfold. First, we introduce Copula Generalized Fiducial Inference (CGFI), a new likelihood-free confidence distribution framework for estimating-equation models. By treating the empirical estimating equations as the stochastic object underlying inference, CGFI extends generalized fiducial methodology beyond models with explicit data-generating mechanisms.

Second, we establish theoretical properties of the proposed construction. Under standard regularity conditions for estimating equations, we show that the CGFI distribution is asymptotically centered at the true parameter value and that its asymptotic covariance structure agrees with the sampling distribution of the corresponding estimating-equation estimator. These results provide the theoretical foundation for interpreting CGFI draws as a frequentist confidence distribution.

Third, we study the finite-sample behavior of CGFI through simulations involving nonlinear estimating equations, increasing numbers of moment conditions, and dependent estimating functions. The results demonstrate that incorporating the estimated dependence structure among moment conditions improves uncertainty quantification relative to procedures based on independent perturbations or standard asymptotic approximations.

Fourth, the proposed framework provides a foundation for extending fiducial inference to broader classes of modern statistical procedures. Because many estimators used in statistics and econometrics are naturally expressed through estimating equations rather than likelihood functions, CGFI offers a general mechanism for constructing likelihood-free confidence distributions in complex models.

The remainder of the paper is organized as follows. Section 2 introduces the CGFI construction and describes the copula-based perturbation mechanism. Section 3 establishes the theoretical properties of the resulting confidence distribution. Section 4 evaluates finite-sample performance through simulation studies. Section 5 concludes and discusses directions for future research.

2 Copula-Based Generalized Fiducial Inference (CGFI) for Estimating Equations

2.1 CGFI as a Generalized Fiducial Construction

The central component of generalized fiducial inference is the inversion of a stochastic equation that links an observed quantity to an unknown parameter and an auxiliary random variable. In classical settings, this representation is written as

$$X = G(U, \theta),$$

where the distribution of U is known. The fiducial distribution is obtained by treating the observed data as fixed and propagating the uncertainty associated with the auxiliary variable through the inverse mapping.

For estimating-equation models, an analogous stochastic representation is obtained by considering the empirical estimating equation

$$\bar{g}_n(\theta) = \frac{1}{n} \sum_{i=1}^n g(X_i, \theta).$$

At the true parameter value θ_0 , the empirical estimating equation satisfies

$$\sqrt{n}\bar{g}_n(\theta_0) \xrightarrow{d} N(0, \Sigma),$$

where

$$\Sigma = \text{Var}\{g(X, \theta_0)\}$$

is the covariance matrix of the estimating functions.

CGFI interprets this asymptotic stochastic representation as the analogue of the auxiliary variable in generalized fiducial inference. Specifically, let

$$\eta \sim N(0, \hat{\Sigma}),$$

where $\hat{\Sigma}$ is an estimate of the covariance matrix of the empirical estimating equations. A fiducial draw θ^* is then obtained by solving

$$\sqrt{n}\bar{g}_n(\theta^*) + \eta = 0.$$

Repeated realizations of η generate a distribution over the parameter space. This distribution is interpreted as a confidence distribution obtained by inverting the stochastic behavior of the estimating equations.

The key distinction from conventional resampling methods is that the random perturbation is not used to approximate the sampling distribution of an estimator. Instead, the perturbed estimating equation defines a stochastic inverse problem, and the resulting distribution is directly constructed on the parameter space. Thus, CGFI extends the generalized fiducial principle to settings where the parameter is characterized through moment restrictions rather than an explicit data-generating equation.

2.2 The CGFI construction

Let θ_0 denote the true parameter satisfying

$$E\{g(X, \theta_0)\} = 0.$$

The empirical estimating equations evaluated at the true parameter satisfy

$$\sqrt{n}\bar{g}_n(\theta_0) \xrightarrow{d} N(0, \Sigma),$$

where

$$\Sigma = \text{Var}\{g(X, \theta_0)\} = E[g(X, \theta_0)g(X, \theta_0)^\top].$$

Thus, the sampling uncertainty of the empirical estimating equations provides a natural stochastic object for fiducial inversion.

Let $\hat{\Sigma}$ denote a consistent estimator of Σ . We decompose the estimated covariance matrix as

$$\hat{\Sigma} = \hat{D}\hat{R}\hat{D},$$

where $\hat{D}$ is a diagonal matrix containing the estimated marginal standard deviations and $\hat{R}$ is the estimated correlation matrix. To generate a stochastic perturbation of the empirical estimating equations, we use a Gaussian copula representation. Specifically, let

$$U^* = (U_1^*, \dots, U_q^*)^\top$$

be a random vector with Gaussian copula dependence parameter $\hat{R}$. Define

$$Z_j^* = \Phi^{-1}(U_j^*), \quad j = 1, \dots, q,$$

where Φ^{-1} denotes the inverse standard normal distribution function. The simulated perturbation is then constructed as

$$\eta^* = \hat{D}Z^*.$$

This representation separates the marginal variability of the estimating equations from their dependence structure. Although this paper focuses on the Gaussian copula construction, the framework separates dependence modeling from marginal specification and can accommodate alternative copula families or marginal distributions in future extensions.

Given a realization η^* , the CGFI parameter draw θ^* is defined as the solution to the perturbed estimating equation criterion

$$\theta^* = \arg \min_{\theta} \left\{ \sqrt{n}\bar{g}_n(\theta) + \eta^* \right\}^\top W_n \left\{ \sqrt{n}\bar{g}_n(\theta) + \eta^* \right\},$$

where W_n is a positive definite weighting matrix. A natural choice is the continuously updated GMM weighting matrix

$$W_n = \hat{\Sigma}^{-1},$$

although other positive definite weighting matrices may be used depending on the estimating-equation model. This formulation accommodates both exactly identified and overidentified estimating equation models. In the exactly identified case, the criterion is minimized when

$$\sqrt{n}\bar{g}_n(\theta^*) + \eta^* = 0,$$

recovering direct inversion of the perturbed estimating equations. In the overidentified case, the construction corresponds to inversion through a perturbed generalized method of moments criterion.

Repeating this procedure for independent realizations

$$\eta_1^*, \eta_2^*, \dots, \eta_M^*$$

produces draws

$$\theta_1^*, \theta_2^*, \dots, \theta_M^*$$

from the CGFI distribution.

The CGFI construction parallels generalized fiducial inference while changing the fundamental object being inverted. Classical GFI obtains parameter uncertainty through

$$X = G(U, \theta) \implies \theta = G^{-1}(X, U),$$

where randomness enters through an auxiliary variable generating the observations. In contrast, CGFI uses

$$\sqrt{n}\bar{g}_n(\theta) + \eta^* = 0$$

as the underlying stochastic representation, where randomness enters through the sampling distribution of the empirical estimating equations. Thus, CGFI replaces the unavailable structural equation for the observations with a stochastic representation of the estimating equations themselves.

This distinction is important because estimating-equation models often contain all information required for inference without admitting a convenient likelihood or structural representation. By treating the empirical estimating equations as the fundamental inferential object, CGFI extends generalized fiducial inference to a broad class of models in which uncertainty is naturally characterized through moment restrictions rather than a fully specified probability model.

3 Formal Construction of the CGFI Distribution

3.1 Definition of the CGFI confidence distribution

The CGFI procedure constructs a distribution for the unknown parameter by propagating the sampling uncertainty of the empirical estimating equations through a generalized fiducial inversion. The construction begins with the asymptotic representation

$$\sqrt{n}\bar{g}_n(\theta_0) \xrightarrow{d} N(0, \Sigma),$$

which motivates the stochastic perturbation of the empirical estimating equations introduced in Section 2.

Let $\mathcal{E}_n^*$ denote the perturbed empirical estimating equation,

$$\mathcal{E}_n^*(\theta) = \sqrt{n}\bar{g}_n(\theta) + \eta^*,$$

where η^* is generated according to the Gaussian copula construction described previously. The corresponding perturbed GMM criterion is

$$Q_n^*(\theta) = \mathcal{E}_n^*(\theta)^\top W_n \mathcal{E}_n^*(\theta).$$

For each realization of η^* , define the CGFI parameter draw as

$$\theta^* = \arg \min_{\theta \in \Theta} Q_n^*(\theta).$$

The randomness in θ^* therefore arises entirely from the simulated perturbation of the empirical estimating equations, while the observed data enter through $\bar{g}_n(\theta)$.

The CGFI confidence distribution is defined as the distribution induced by the random variable θ^* conditional on the observed sample. Specifically, let

$$H_n^{\text{CGFI}}(\theta) = P^*(\theta^* \leq \theta \mid X_1, \dots, X_n),$$

where P^* denotes probability with respect to the simulation distribution of η^* . This distribution provides a complete probabilistic representation of uncertainty about θ_0 and can be used to construct confidence intervals and other inferential procedures. For example, a $(1-\alpha)$ CGFI confidence interval for the j th component of θ is given by

$$\left[q_{\alpha/2,j}^*, q_{1-\alpha/2,j}^* \right],$$

where $q_{\gamma,j}^*$ denotes the γ quantile of the marginal CGFI distribution for θ_j^* .

3.2 Relationship with generalized fiducial inference

The CGFI construction can be viewed as a generalized fiducial procedure in which the source of randomness is shifted from the data-generating equation to the estimating equations defining the parameter. Classical generalized fiducial inference begins with a stochastic representation of the form

$$X = G(U, \theta),$$

where the auxiliary variable U generates the observations. Inverting this relationship yields

$$\theta = G^{-1}(X, U),$$

and the distribution of the inverted parameter values defines the fiducial distribution.

For estimating-equation models, such a representation is generally unavailable. Instead, CGFI uses the asymptotic stochastic representation

$$\sqrt{n}\bar{g}_n(\theta_0) \approx \eta,$$

where

$$\eta \sim N(0, \Sigma).$$

The fiducial inversion is therefore performed through the perturbed estimating equation criterion

$$\theta^* = \arg \min_{\theta} \left\{ \sqrt{n}\bar{g}_n(\theta) + \eta^* \right\}^\top W_n \left\{ \sqrt{n}\bar{g}_n(\theta) + \eta^* \right\}.$$

This formulation preserves the central idea of generalized fiducial inference: uncertainty is generated by reversing a stochastic relationship that links observable quantities to unknown parameters. The key difference is that CGFI replaces the unavailable structural equation for the observations with a stochastic representation of the empirical estimating equations. Consequently, the method extends the fiducial principle to models where parameters are defined through moment restrictions rather than a fully specified probability model.

Remark 1. *Although the CGFI construction uses a perturbed GMM criterion in overidentified models, the objective differs from conventional GMM inference. The perturbation is not used to estimate the sampling distribution of the estimator; rather, it defines a stochastic inverse problem whose solution induces a distribution directly on the parameter space.*

3.3 Computational implementation

In practice, the CGFI distribution is approximated through Monte Carlo simulation. Let M denote the number of simulation draws. The algorithm proceeds as follows.

1. Compute the sample estimating equations $\bar{g}_n(\theta)$ and obtain consistent estimates $\hat{\Sigma}$ and W_n .
2. Decompose the covariance estimator as

$$\hat{\Sigma} = \hat{D}\hat{R}\hat{D}.$$

3. For $m = 1, \dots, M$, generate a Gaussian copula realization

$$U_m^* \sim C_{\hat{R}},$$

transform the simulated uniforms using

$$Z_{mj}^* = \Phi^{-1}(U_{mj}^*),$$

and construct

$$\eta_m^* = \hat{D}Z_m^*.$$

4. Obtain the corresponding parameter draw

$$\theta_m^* = \arg \min_{\theta} \left\{ \sqrt{n} \bar{g}_n(\theta) + \eta_m^* \right\}^\top W_n \left\{ \sqrt{n} \bar{g}_n(\theta) + \eta_m^* \right\}.$$

5. Approximate the CGFI confidence distribution using the empirical distribution of

$$\theta_1^*, \dots, \theta_M^*.$$

For clarity, the computational steps of the CGFI procedure are summarized in Algorithm 1.

The computational procedure requires repeated optimization of a perturbed estimating equation criterion rather than repeated estimation of the model on bootstrap samples. This distinction can substantially reduce computational burden in settings where estimating equations are inexpensive to evaluate but repeated resampling and model re-estimation are costly.

The following section establishes the theoretical properties of CGFI. In particular, we show that the resulting confidence distribution consistently concentrates around the true parameter, derive a Bernstein–von Mises result describing its asymptotic normality, and establish higher-order properties of the resulting fiducial approximation.

The following section establishes the theoretical properties of CGFI.

4 Theoretical Properties of CGFI

4.1 Regularity Conditions

This section establishes the asymptotic properties of the CGFI distribution. The results are developed for a general class of estimating-equation models, including both exactly identified and overidentified settings. Let

$$\theta_0 \in \Theta \subset \mathbb{R}^p$$

Algorithm 1 Copula Generalized Fiducial Inference (CGFI)

- 1: **Input:** Estimating function $g(X, \theta)$, observed data $\{X_i\}_{i=1}^n$, number of fiducial draws M .
- 2: Compute the empirical estimating equation

$$\bar{g}_n(\theta) = \frac{1}{n} \sum_{i=1}^n g(X_i, \theta)$$

and estimate the covariance matrix $\hat{\Sigma}$.

- 3: Decompose the covariance estimator as

$$\hat{\Sigma} = \hat{D}\hat{R}\hat{D}.$$

- 4: For $m = 1, \dots, M$, generate a Gaussian copula draw with dependence matrix $\hat{R}$ and obtain

$$\eta_m^* = \hat{D}Z_m^*.$$

- 5: Compute the CGFI draw

$$\theta_m^* = \arg \min_{\theta} \left\{ \sqrt{n}\bar{g}_n(\theta) + \eta_m^* \right\}^\top W_n \left\{ \sqrt{n}\bar{g}_n(\theta) + \eta_m^* \right\}.$$

- 6: Use the empirical distribution of

$$\{\theta_1^*, \dots, \theta_M^*\}$$

to construct CGFI confidence intervals.

- 7: **Output:** CGFI confidence distribution and confidence intervals.
-

denote the true parameter, where Θ is a compact parameter space, and suppose that θ_0 is characterized by the population moment condition

$$E\{g(X, \theta_0)\} = 0,$$

where

$$g(X, \theta) \in \mathbb{R}^q$$

is an estimating function with $q \geq p$.

Define the population moment function

$$G(\theta) = E\{g(X, \theta)\},$$

and let

$$G_0 = \left. \frac{\partial G(\theta)}{\partial \theta^\top} \right|_{\theta=\theta_0}$$

denote the Jacobian matrix evaluated at the true parameter value. The following assumptions provide sufficient conditions for the theoretical results.

The theoretical objective of this section is to establish that the proposed Copula Generalized Fiducial Inference (CGFI) procedure produces an asymptotically valid confidence distribution for parameters identified through estimating equations. Specifically, we show that the conditional distribution of CGFI draws concentrates around the true parameter and asymptotically reproduces the sampling distribution of the corresponding estimating-equation estimator. The analysis proceeds by characterizing the stochastic behavior of the perturbed estimating equations and showing that the resulting fiducial draws inherit the same first-order asymptotic distribution as the underlying estimating-equation estimator. Consequently, the proposed procedure provides asymptotically valid confidence intervals without requiring specification of a likelihood function or an explicit structural equation for the observed data.

Assumption 1. *The parameter space Θ is compact, and θ_0 is an interior point of Θ .*

Assumption 2. *The population moment condition uniquely identifies the true parameter. Specifically,*

$$G(\theta) = 0$$

if and only if

$$\theta = \theta_0.$$

Assumption 3. *The estimating function $g(X, \theta)$ is continuously differentiable in a neighborhood of θ_0 , and the Jacobian matrix*

$$G_0 = E \left[\frac{\partial g(X, \theta_0)}{\partial \theta^\top} \right]$$

has full column rank.

Assumption 4. *The estimating function satisfies*

$$E\|g(X, \theta_0)\|^{2+\delta} < \infty$$

for some $\delta > 0$, and the covariance matrix

$$\Sigma = \text{Var}\{g(X, \theta_0)\}$$

is positive definite.

Assumption 5. *The derivative of the estimating function satisfies a uniform convergence condition in a neighborhood of θ_0 :*

$$\sup_{\theta \in \Theta} \left\| \frac{\partial \bar{g}_n(\theta)}{\partial \theta^\top} - \frac{\partial G(\theta)}{\partial \theta^\top} \right\| \xrightarrow{p} 0.$$

Assumption 6. *The sample estimating function satisfies a uniform law of large numbers in a neighborhood of θ_0 , such that*

$$\sup_{\theta \in \Theta} \|\bar{g}_n(\theta) - G(\theta)\| \xrightarrow{p} 0,$$

and the estimated covariance matrix satisfies

$$\hat{\Sigma} \xrightarrow{p} \Sigma.$$

Assumption 7. *The estimated correlation matrix obtained from the covariance decomposition*

$$\hat{\Sigma} = \hat{D} \hat{R} \hat{D}$$

satisfies

$$\hat{R} \xrightarrow{p} R.$$

Consequently, the Gaussian copula perturbation consistently estimates the dependence structure of the empirical estimating equations.

Under Assumptions 1–7, the empirical estimating equations satisfy the multivariate central limit theorem,

$$\sqrt{n} \bar{g}_n(\theta_0) \xrightarrow{d} N(0, \Sigma).$$

The Gaussian copula construction used by CGFI provides simulated perturbations that reproduce this limiting dependence structure. Importantly, the Gaussian copula is not imposed as a parametric model for the observed data; rather, it provides a stochastic representation of the asymptotic Gaussian distribution of the empirical estimating equations. Consequently, the simulated moment perturbations serve as an asymptotically valid representation of the sampling uncertainty contained in the estimating equations.

To define the CGFI draws formally, let η^* denote a simulated perturbation generated from the Gaussian copula representation of the estimated covariance matrix. The CGFI draw is defined as the solution to the perturbed estimating equation

$$\sqrt{n} \bar{g}_n(\theta^*) + \eta^* = 0$$

in exactly identified models. For overidentified models, the corresponding GMM-type criterion is used:

$$\theta^* = \arg \min_{\theta} Q_n^*(\theta),$$

where

$$Q_n^*(\theta) = \left\{ \sqrt{n} \bar{g}_n(\theta) + \eta^* \right\}^\top W_n \left\{ \sqrt{n} \bar{g}_n(\theta) + \eta^* \right\},$$

and

$$W_n \xrightarrow{p} W$$

for a positive definite matrix W .

Finally, define

$$V = (G_0^\top W G_0)^{-1} G_0^\top W \Sigma W G_0 (G_0^\top W G_0)^{-1},$$

which is the standard sandwich covariance matrix describing the asymptotic sampling variability of the corresponding estimating-equation estimator.

4.2 Consistency of the CGFI Distribution

The first theoretical result establishes that the CGFI distribution consistently concentrates around the true parameter value. Although CGFI introduces additional randomness through the simulated perturbation η^* , this perturbation is of the same stochastic order as the sampling variability of the empirical estimating equations. Consequently, the perturbation does not affect the probability limit of the resulting fiducial distribution.

Let

$$H_n^{\text{CGFI}}(\cdot)$$

denote the conditional distribution of the CGFI draws given the observed sample. The following theorem shows that this distribution asymptotically degenerates at the true parameter value.

Theorem 4.1 (Consistency of the CGFI distribution). *Suppose Assumptions 1–7 hold. Then, for every $\epsilon > 0$,*

$$P^*(\|\theta^* - \theta_0\| > \epsilon) \xrightarrow{p} 0,$$

where P^* denotes probability with respect to the CGFI simulation distribution conditional on the observed sample.

Equivalently, the CGFI distribution converges weakly to a point mass concentrated at θ_0 .

Proof. Consider the CGFI objective function

$$Q_n^*(\theta) = \{\sqrt{n}\bar{g}_n(\theta) + \eta^*\}^\top W_n \{\sqrt{n}\bar{g}_n(\theta) + \eta^*\}.$$

Multiplying by $1/n$ gives the equivalent criterion

$$\tilde{Q}_n^*(\theta) = \{\bar{g}_n(\theta) + n^{-1/2}\eta^*\}^\top W_n \{\bar{g}_n(\theta) + n^{-1/2}\eta^*\}.$$

By construction of the Gaussian copula perturbation,

$$\eta^* = O_{p^*}(1) \quad \text{in probability.}$$

Therefore,

$$n^{-1/2}\eta^* \xrightarrow{p^*} 0.$$

It follows that the perturbed empirical moment function satisfies

$$\bar{g}_n(\theta) + n^{-1/2}\eta^* = \bar{g}_n(\theta) + o_{p^*}(1)$$

uniformly in $\theta \in \Theta$.

By Assumption 5,

$$\sup_{\theta \in \Theta} \|\bar{g}_n(\theta) - G(\theta)\| \xrightarrow{p} 0.$$

Together with the consistency of the weighting matrix,

$$W_n \xrightarrow{p} W,$$

we obtain

$$\sup_{\theta \in \Theta} \left| \tilde{Q}_n^*(\theta) - G(\theta)^\top W G(\theta) \right| \xrightarrow{p} 0.$$

The limiting objective function is therefore

$$Q(\theta) = G(\theta)^\top W G(\theta).$$

Because W is positive definite and Assumption 2 implies that

$$G(\theta) = 0$$

if and only if

$$\theta = \theta_0,$$

the limiting criterion function has a unique minimizer at θ_0 .

By the argmin consistency theorem,

$$\theta^* = \arg \min_{\theta} Q_n^*(\theta) \xrightarrow{p^*} \theta_0$$

in probability. Equivalently,

$$P^*(\|\theta^* - \theta_0\| > \epsilon) \xrightarrow{p} 0$$

for every $\epsilon > 0$.

Thus, the CGFI distribution consistently concentrates around the true parameter value. $\square$

The consistency result establishes that CGFI provides a valid asymptotic representation of parameter uncertainty. The remaining results characterize the shape of the limiting distribution and show that the conditional distribution of CGFI draws reproduces the first-order sampling distribution of the underlying estimating-equation estimator.

Proposition 4.2 (Validity of the Gaussian copula perturbation). *Suppose Assumptions 1–7 hold. Then the conditional distribution of the CGFI perturbation satisfies*

$$\eta^* \xrightarrow{d^*} N(0, \Sigma)$$

in probability. Specifically, the Gaussian copula construction based on

$$\widehat{\Sigma} = \widehat{D}\widehat{R}\widehat{D}$$

consistently estimates the limiting joint distribution of the empirical estimating equations.

Proof. By construction,

$$\eta^* = \widehat{D}Z^*,$$

where Z^* is generated from a Gaussian copula with correlation matrix $\widehat{R}$.

By Assumption 7,

$$\widehat{R} \xrightarrow{p} R,$$

and by Assumption 6,

$$\widehat{D} \xrightarrow{p} D.$$

Therefore, the Gaussian copula transformation converges to the limiting Gaussian dependence structure characterized by

$$\Sigma = DRD.$$

Hence,

$$\eta^* \xrightarrow{d^*} N(0, \Sigma)$$

in probability by the continuous mapping theorem. $\square$

4.3 Asymptotic Normality and Bernstein–von Mises-Type Property of CGFI

The consistency result establishes that the CGFI distribution concentrates around the true parameter value. We next characterize the asymptotic shape of the CGFI distribution. In particular, we show that the conditional distribution of CGFI draws, given the observed sample, asymptotically reproduces the sampling distribution of the corresponding estimating-equation estimator.

Let

$$\hat{\theta} = \arg \min_{\theta} \bar{g}_n(\theta)^\top W_n \bar{g}_n(\theta)$$

denote the conventional estimating-equation estimator. Under Assumptions 1–7,

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, V),$$

where

$$V = (G_0^\top W G_0)^{-1} G_0^\top W \Sigma W G_0 (G_0^\top W G_0)^{-1}$$

is the standard sandwich covariance matrix.

The following theorem establishes that CGFI provides an asymptotically valid confidence distribution by showing that the conditional distribution of CGFI draws converges to the same Gaussian limit.

Theorem 4.3 (Asymptotic normality of the CGFI distribution). *Suppose Assumptions 1–7 hold. Additionally, suppose that the estimating function satisfies the conditions required for a first-order asymptotic expansion of the estimating-equation estimator. Then,*

$$\sup_{A \in \mathcal{B}(\mathbb{R}^p)} \left| P^* \left(\sqrt{n}(\theta^* - \hat{\theta}) \in A \right) - \Phi_V(A) \right| \xrightarrow{p} 0,$$

where P^* denotes probability with respect to the CGFI simulation distribution conditional on the observed sample, and Φ_V denotes the distribution function of a multivariate normal random variable with mean zero and covariance matrix V .

Equivalently, the CGFI confidence distribution satisfies

$$H_n^{\text{CGFI}} \left(\hat{\theta} + \frac{t}{\sqrt{n}} \right) \xrightarrow{p} \Phi_V(t)$$

uniformly in $t \in \mathbb{R}^p$.

Therefore, the CGFI distribution is asymptotically equivalent to the sampling distribution of the corresponding estimating-equation estimator. This result provides a Bernstein–von Mises-type property for the CGFI confidence distribution: conditional on the observed data, the fiducial distribution asymptotically matches the frequentist sampling distribution of the estimator.

Proof. The CGFI draw is defined as the minimizer of

$$Q_n^*(\theta) = \left\{ \sqrt{n} \bar{g}_n(\theta) + \eta^* \right\}^\top W_n \left\{ \sqrt{n} \bar{g}_n(\theta) + \eta^* \right\}.$$

The first-order condition is

$$G_n(\theta^*)^\top W_n \left\{ \sqrt{n} \bar{g}_n(\theta^*) + \eta^* \right\} = 0,$$

where

$$G_n(\theta) = \frac{\partial \bar{g}_n(\theta)}{\partial \theta^\top}.$$

The conventional estimator satisfies

$$G_n(\hat{\theta})^\top W_n \sqrt{n} \bar{g}_n(\hat{\theta}) = 0.$$

Expanding the first-order condition for θ^* around $\hat{\theta}$ gives

$$0 = G_n(\hat{\theta})^\top W_n \eta^* + G_n(\hat{\theta})^\top W_n G_n(\hat{\theta}) \sqrt{n}(\theta^* - \hat{\theta}) + o_{p^*}(1).$$

Therefore,

$$\sqrt{n}(\theta^* - \hat{\theta}) = - \left[G_n(\hat{\theta})^\top W_n G_n(\hat{\theta}) \right]^{-1} G_n(\hat{\theta})^\top W_n \eta^* + o_{p^*}(1).$$

By consistency of the estimator and the Jacobian,

$$G_n(\hat{\theta}) \xrightarrow{p} G_0,$$

and by Assumption 6,

$$W_n \xrightarrow{p} W.$$

Hence,

$$\sqrt{n}(\theta^* - \hat{\theta}) = -(G_0^\top W G_0)^{-1} G_0^\top W \eta^* + o_{p^*}(1).$$

From Proposition 1,

$$\eta^* \xrightarrow{d^*} N(0, \Sigma)$$

in probability. Applying the continuous mapping theorem,

$$-(G_0^\top W G_0)^{-1} G_0^\top W \eta^* \xrightarrow{d^*} N(0, V),$$

where

$$V = (G_0^\top W G_0)^{-1} G_0^\top W \Sigma W G_0 (G_0^\top W G_0)^{-1}.$$

Therefore,

$$\sqrt{n}(\theta^* - \hat{\theta}) \xrightarrow{d^*} N(0, V)$$

in probability, which establishes the asymptotic normality of the CGFI confidence distribution. $\square$

The asymptotic normality result implies that the quantiles of the CGFI confidence distribution provide asymptotically valid confidence limits. The following corollary formalizes the resulting frequentist coverage property.

Corollary 4.4 (Asymptotic coverage of CGFI intervals). *Under the conditions of Theorem 2, let*

$$C_{n,\alpha} = \left[q_{\alpha/2}^*, q_{1-\alpha/2}^* \right]$$

denote the marginal CGFI confidence interval for a component of θ_0 . Then

$$P(\theta_{0j} \in C_{n,\alpha}) \longrightarrow 1 - \alpha.$$

This result establishes that CGFI confidence intervals achieve nominal frequentist coverage asymptotically. Thus, although CGFI constructs uncertainty through stochastic inversion rather than repeated-sampling approximation of an estimator, the resulting confidence distribution provides valid frequentist inference.

4.4 First-Order Expansion and Extensions

The Bernstein–von Mises-type result establishes the first-order asymptotic validity of CGFI by showing that the conditional distribution of fiducial draws reproduces the limiting sampling distribution of the corresponding estimating-equation estimator. The result also provides a foundation for studying higher-order refinements of CGFI confidence procedures.

To see the source of higher-order behavior, consider a local expansion of the CGFI draw around the conventional estimator. Under additional smoothness conditions, a Taylor expansion of the perturbed estimating equation gives the representation

$$\sqrt{n}(\theta^* - \hat{\theta}) = -A_0\eta^* + o_{p^*}(1),$$

where

$$A_0 = (G_0^\top W G_0)^{-1} G_0^\top W.$$

The leading term captures the asymptotic uncertainty generated by the empirical estimating equations. Higher-order corrections arise from nonlinearities in the estimating equations, estimation of the Jacobian matrix, uncertainty in the weighting matrix, and approximation error arising from estimating the perturbation distribution.

A complete higher-order theory for CGFI would require additional assumptions controlling the interaction between nonlinear estimating equations, estimation of the dependence structure, and approximation error introduced by the copula perturbation mechanism. Because the primary objective of this paper is to establish a general first-order fiducial framework for estimating-equation models, we leave the development of higher-order refinements, including possible Edgeworth expansions and improved finite-sample corrections, for future research.

The first-order theory developed in this paper establishes the primary theoretical guarantee required for CGFI: the resulting confidence distribution is asymptotically equivalent to the sampling distribution of the underlying estimating-equation estimator.

5 Simulation Study

This section evaluates the finite-sample performance of the proposed Copula Generalized Fiducial Inference (CGFI) procedure for estimating-equation models. The simulation study examines four aspects of the proposed method. First, we evaluate performance in high-dimensional estimating-equation systems where the number of moment conditions is large relative to the sample size.

Second, we examine how inference changes as the amount of information increases through larger samples and additional moment conditions. Third, we evaluate robustness to departures from the baseline design, including dependent moment conditions, non-Gaussian disturbances, and heteroskedasticity. Finally, we compare the computational cost of CGFI with conventional bootstrap inference.

Throughout the simulations, confidence intervals are constructed using three methods:

1. the conventional Wald approximation based on the asymptotic covariance matrix,
2. the nonparametric bootstrap, and
3. the proposed CGFI procedure.

Unless otherwise specified, all experiments use 100 Monte Carlo replications and 200 draws for each simulation-based confidence procedure. Unless otherwise specified, all experiments use 100 Monte Carlo replications and 200 draws for each simulation-based confidence procedure. Coverage probabilities are evaluated relative to the nominal 95 percent confidence level, and Monte Carlo standard errors are reported to quantify simulation uncertainty.

5.1 Data Generating Process

The baseline data generating process is based on the nonlinear estimating equation model

$$Y_i = \theta_0 X_i + \frac{1}{2} X_i^2 + u_i,$$

where the parameter of interest is

$$\theta_0 = 2.$$

The regressor is generated independently according to

$$X_i \sim N(0, 1),$$

and the baseline disturbance follows a Student- t distribution,

$$u_i \sim t_5.$$

The heavy-tailed disturbance introduces non-Gaussian sampling variation while preserving the moment restrictions required for estimation. The nonlinear quadratic component ensures that the estimating equations are not equivalent to a simple linear regression problem.

Identification is obtained through the estimating equations

$$E \left[Z_i \left(Y_i - \theta X_i - \frac{1}{2} X_i^2 \right) \right] = 0,$$

where

$$Z_i = (Z_{i1}, \dots, Z_{iq})^\top$$

is a vector of moment functions. In the baseline design, instruments are independent and generated as

$$Z_{ij} \sim N(0, 1).$$

This specification is intentionally designed to isolate the behavior of the confidence procedures as the dimension of the estimating-equation system increases. The moment functions therefore provide multiple estimating equations without introducing additional identification complications.

The corresponding empirical estimating function is

$$\bar{g}_n(\theta) = \frac{1}{n} \sum_{i=1}^n Z_i \left(Y_i - \theta X_i - \frac{1}{2} X_i^2 \right).$$

The estimator is obtained using the two-step nonlinear GMM criterion,

$$\hat{\theta} = \arg \min_{\theta} \bar{g}_n(\theta)^\top W_n \bar{g}_n(\theta),$$

where W_n is the estimated optimal weighting matrix.

For CGFI, uncertainty is generated by perturbing the empirical estimating equations through simulated moment shocks,

$$\sqrt{n} \bar{g}_n(\theta) + \eta^* = 0,$$

where η^* is sampled from the Gaussian copula approximation to the joint distribution of the empirical estimating equations, preserving the estimated dependence structure across moments. The resulting collection of solutions defines the CGFI confidence distribution.

The baseline design uses

$$n = 500, \quad q = 100,$$

unless otherwise noted. Subsequent experiments modify one component of the data generating process at a time to examine the robustness of the proposed procedure.

5.2 Increasing the Number of Moment Conditions

The first experiment examines the behavior of the competing inference procedures as the dimension of the estimating-equation system increases. This setting is important because many modern applications of estimating equations involve a large number of moment restrictions, where first-order asymptotic approximations may provide inaccurate uncertainty quantification.

The sample size is fixed at

$$n = 500,$$

while the number of moment conditions varies according to

$$q \in \{10, 50, 100, 250\}.$$

For each value of q , the model is estimated using the same nonlinear GMM procedure, and confidence intervals are constructed using Wald, bootstrap, and CGFI inference.

Table 2 reports the simulation results. When the number of moment conditions is small ($q = 10$), all procedures provide coverage close to the nominal 95 percent level. However, as the number of moments increases, the performance of the Wald approximation deteriorates substantially. In particular, when $q = 250$, Wald coverage falls to 60 percent despite producing the shortest confidence intervals. This pattern indicates that first-order asymptotic approximations may become less reliable in finite samples when the dimension of the estimating-equation system becomes large relative to the sample size.

Bootstrap inference provides coverage close to or slightly above the nominal level throughout the experiment, with coverage between 99 and 100 percent. However, bootstrap intervals are generally wider than those obtained from Wald inference and require substantially greater computational effort. The computational burden becomes particularly noticeable when $q = 250$, where each bootstrap replication requires repeated nonlinear GMM estimation.

CGFI maintains coverage closer to the nominal level than the Wald procedure across all values of q . Coverage remains between 96 and 98 percent when the number of moments exceeds 100, while confidence intervals remain comparable in length to bootstrap intervals. These results suggest that the Gaussian copula perturbation provides a useful approximation to the finite-sample distribution of the estimating equations, even when the number of moment conditions is large.

Table 2: Simulation Results: Increasing Number of Moment Conditions

q	Method	Coverage	MCSE	Avg. Length	Avg. Time (sec)
10	Wald	0.97	0.017	1.658	—
10	Bootstrap	1.00	0.000	1.500	0.295
10	CGFI	0.98	0.014	1.592	0.039
50	Wald	0.92	0.027	0.622	—
50	Bootstrap	1.00	0.000	0.636	1.411
50	CGFI	0.96	0.020	0.659	0.063
100	Wald	0.92	0.027	0.387	—
100	Bootstrap	1.00	0.000	0.475	4.684
100	CGFI	0.97	0.017	0.497	0.164
250	Wald	0.60	0.049	0.175	—
250	Bootstrap	0.99	0.010	0.399	30.966
250	CGFI	0.98	0.014	0.517	0.927

Figure 1 presents coverage probabilities as the number of moment conditions increases. The figure highlights the deterioration of Wald inference in high-dimensional estimating-equation systems, while CGFI remains substantially closer to the nominal confidence level with considerably lower computational cost than the bootstrap.

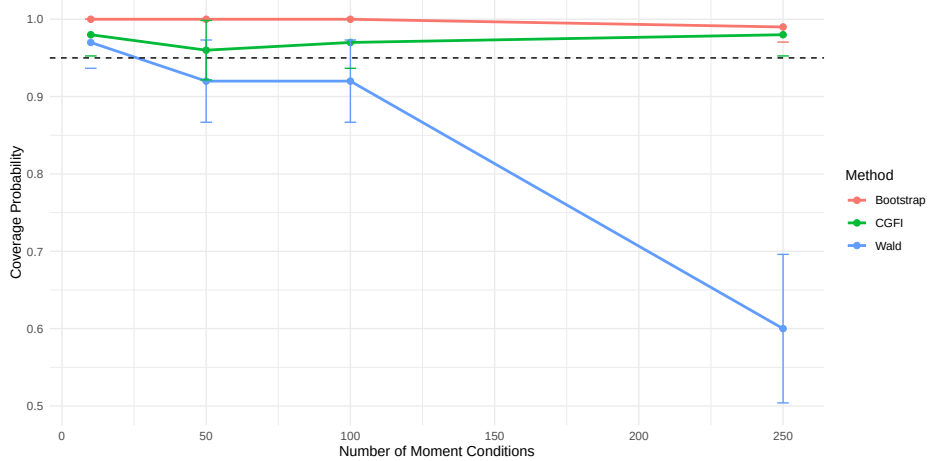


Figure 1: Coverage Probability as the Number of Moment Conditions Increases. The dashed line indicates the nominal 95 percent coverage level.

Figure 2 displays average confidence interval lengths. Wald intervals become increasingly narrow as the number of moments increases, despite their declining coverage. In contrast, CGFI produces intervals of similar magnitude to bootstrap intervals, suggesting that the proposed procedure better captures finite-sample uncertainty.

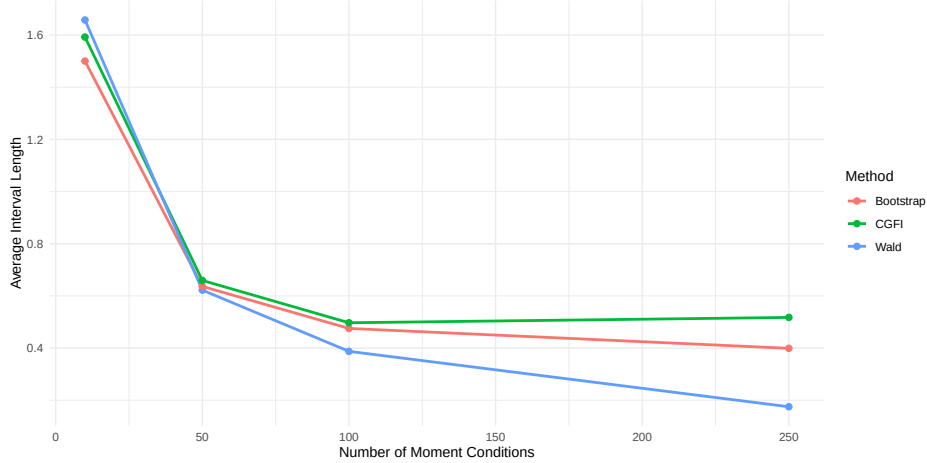


Figure 2: Average Confidence Interval Length as the Number of Moment Conditions Increases.

5.3 Increasing Sample Size

The second experiment evaluates the behavior of CGFI as the amount of information increases. Rather than increasing the sample size while holding the number of moment conditions fixed, we allow the dimension of the estimating-equation system to increase proportionally with the sample size. This design reflects applications where additional observations are often accompanied by an expansion of the available moment conditions.

Specifically, the ratio of moment conditions to observations is held fixed at

$$\frac{q}{n} = 0.2.$$

The sample sizes considered are

$$n \in \{100, 250, 500, 1000\},$$

with corresponding numbers of moments

$$q \in \{20, 50, 100, 200\}.$$

Table 3 reports the results. Across all sample sizes, the Wald approximation exhibits noticeable undercoverage, with coverage ranging from 83 to 90 percent. Although increasing the sample size improves the accuracy of the estimator, the simultaneous increase in the number of moment conditions creates a more challenging inference problem for first-order asymptotic approximations.

CGFI generally provides coverage closer to the nominal 95 percent level than the Wald approximation throughout the experiment. Coverage ranges from 88 to 95 percent, with the largest sample size achieving nominal coverage. These results suggest that CGFI improves finite-sample calibration when both the sample size and the dimension of the estimating-equation system increase.

Bootstrap inference achieves the highest coverage, reaching 100 percent in each specification. However, this improvement comes with wider intervals and substantially greater computational cost. As the sample size and number of moments increase, bootstrap inference requires repeatedly solving the nonlinear GMM optimization problem on resampled datasets.

Table 3: Simulation Results: Increasing Sample Size with Fixed Moment Ratio

n	q	Method	Coverage	MCSE	Avg. Length	Avg. Time (sec)
100	20	Wald	0.90	0.030	0.925	—
100	20	Bootstrap	1.00	0.000	1.176	0.284
100	20	CGFI	0.95	0.022	1.166	0.050
250	50	Wald	0.83	0.038	0.545	—
250	50	Bootstrap	1.00	0.000	0.697	1.169
250	50	CGFI	0.90	0.030	0.706	0.090
500	100	Wald	0.83	0.038	0.379	—
500	100	Bootstrap	1.00	0.000	0.472	5.215
500	100	CGFI	0.88	0.032	0.490	0.193
1000	200	Wald	0.87	0.034	0.271	—
1000	200	Bootstrap	1.00	0.000	0.333	27.448
1000	200	CGFI	0.95	0.022	0.355	0.504

Overall, the first two experiments demonstrate that CGFI provides a useful alternative to conventional asymptotic inference in settings with many estimating equations. Unlike Wald inference, CGFI maintains substantially better coverage as the dimension of the moment system increases. Unlike bootstrap inference, CGFI achieves comparable coverage while requiring a fraction of the computational cost.

5.4 Robustness to Dependent Moment Conditions

The next experiment examines the effect of dependence among the estimating equations. This experiment directly evaluates the role of the Gaussian copula construction in CGFI, which incorporates the empirical dependence structure of the moment conditions into the perturbation distribution.

The instruments are generated from an equicorrelated Gaussian distribution with covariance matrix

$$\Sigma_Z = \begin{pmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \cdots & 1 \end{pmatrix}.$$

We consider dependence levels

$$\rho \in \{0, 0.25, 0.50, 0.75\}.$$

For each design, we simulate 100 datasets with $n = 500$ observations and $q = 100$ moment conditions.

Table 4 reports the results. The Wald procedure exhibits substantial variation in coverage across dependence structures, ranging from 80 to 93 percent. Bootstrap inference remains conservative, achieving coverage near or above the nominal level, but requires substantially greater computation.

CGFI maintains coverage close to the nominal 95 percent level across all dependence specifications, with coverage ranging from 92 to 99 percent. In addition, CGFI automatically adapts its uncertainty quantification as the dependence among moment conditions increases. The increase in interval length for larger values of ρ reflects the additional uncertainty associated with more highly correlated estimating equations.

Table 4: Simulation Results Under Dependent Moment Conditions

ρ	Method	Coverage	MCSE	Avg. Length	Avg. Time (sec)
0.00	Wald	0.85	0.036	0.386	—
0.00	Bootstrap	0.99	0.010	0.481	5.224
0.00	CGFI	0.92	0.027	0.490	0.184
0.25	Wald	0.80	0.040	0.383	—
0.25	Bootstrap	1.00	0.000	0.502	11.110
0.25	CGFI	0.96	0.020	0.613	0.193
0.50	Wald	0.93	0.026	0.391	—
0.50	Bootstrap	1.00	0.000	0.547	14.655
0.50	CGFI	0.99	0.010	0.744	0.196
0.75	Wald	0.83	0.038	0.388	—
0.75	Bootstrap	1.00	0.000	0.579	14.925
0.75	CGFI	0.98	0.014	1.037	0.204

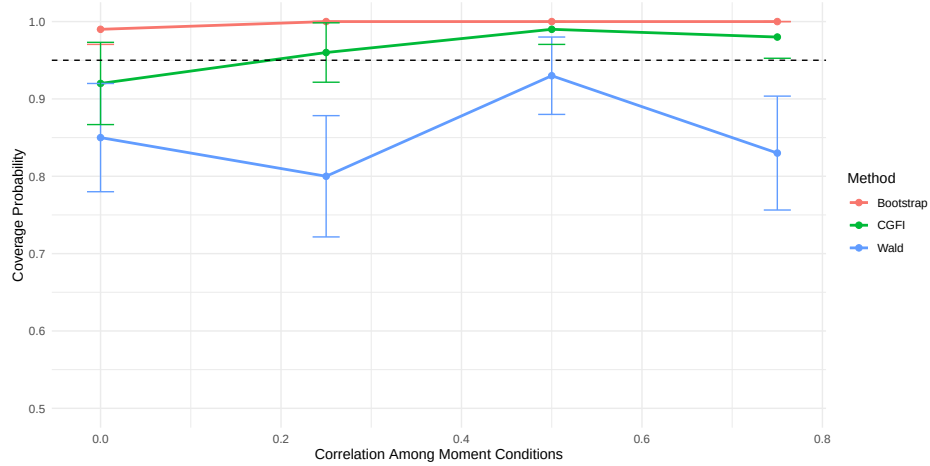


Figure 3: Coverage probability under dependent moment conditions. The dashed line indicates the nominal 95 percent coverage level.

5.5 Robustness to Alternative Error Distributions

The next experiment examines whether the performance of CGFI depends on the distributional assumptions governing the structural disturbance. We consider three error distributions:

$$u_i \sim N(0, 1),$$

$$u_i \sim t_5,$$

and

$$u_i \sim t_3.$$

The latter two specifications introduce increasingly heavy tails. The remaining components of the data generating process are unchanged.

Table 5 reports the results. Under Gaussian errors, CGFI and Wald inference exhibit similar coverage, although both procedures under-cover in this finite-sample design. As the tails of the error distribution become heavier, CGFI provides improved calibration relative to the Wald approximation. However, as the tails of the error distribution become heavier, CGFI provides improved coverage relative to Wald inference. Under t_3 errors, CGFI achieves 98 percent coverage compared with 91 percent for the Wald procedure.

Bootstrap inference remains conservative across all specifications, achieving coverage close to or above the nominal level. However, this improvement comes at the cost of substantially greater computational time.

Error	Method	Coverage	MCSE	Avg. Length	Avg. Time (sec)
Normal	Wald	0.83	0.038	0.331	—
Normal	Bootstrap	0.99	0.010	0.428	4.216
Normal	CGFI	0.83	0.038	0.378	0.154
t_5	Wald	0.85	0.036	0.387	—
t_5	Bootstrap	0.99	0.010	0.487	4.437
t_5	CGFI	0.92	0.027	0.506	0.166
t_3	Wald	0.91	0.029	0.419	—
t_3	Bootstrap	1.00	0.000	0.504	6.976
t_3	CGFI	0.98	0.014	0.659	0.175

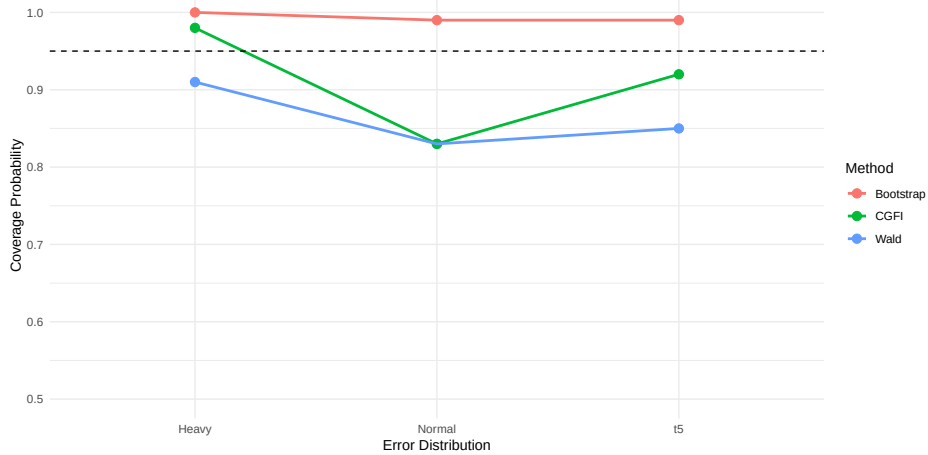


Figure 4: Coverage probability under alternative error distributions. The dashed line indicates the nominal 95 percent coverage level.

5.6 Robustness to Heteroskedastic Disturbances

The following experiment evaluates robustness to heteroskedastic errors. The disturbance is generated as

$$u_i = \sigma(X_i)\epsilon_i,$$

where

$$\epsilon_i \sim t_5,$$

and the variance function $\sigma(X_i)$ is modified to generate increasing levels of heteroskedasticity. We consider homoskedastic, moderate, and strong heteroskedastic designs.

Table 6 reports the results. Wald inference exhibits persistent undercoverage, ranging from 83 to 89 percent. Bootstrap inference remains conservative, while CGFI provides coverage closer to the nominal level. In particular, under strong heteroskedasticity, CGFI achieves 97 percent coverage.

These results suggest that perturbing the empirical estimating equations allows CGFI to incorporate the observed variability of the moment conditions without requiring an explicit parametric specification of the heteroskedastic variance function.

Table 6: Simulation Results: Heteroskedastic Disturbances

Error Structure	Method	Coverage	MCSE	Avg. Length
Homoskedastic	Wald	0.88	0.032	0.378
Homoskedastic	Bootstrap	1.00	0.000	0.471
Homoskedastic	CGFI	0.92	0.027	0.482
Moderate	Wald	0.83	0.038	0.543
Moderate	Bootstrap	1.00	0.000	0.763
Moderate	CGFI	0.91	0.029	0.725
Strong	Wald	0.89	0.031	0.427
Strong	Bootstrap	1.00	0.000	0.479
Strong	CGFI	0.97	0.017	0.722

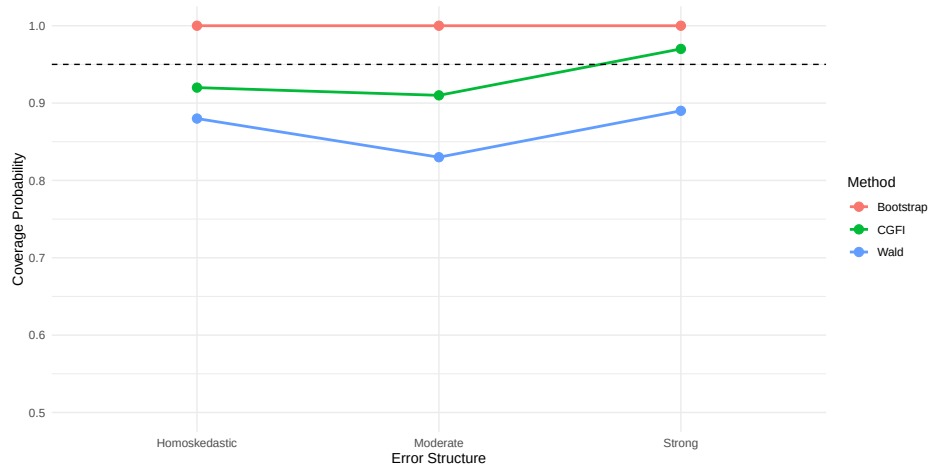


Figure 5: Coverage probability under heteroskedastic disturbances.

5.7 Summary of Simulation Findings

Across the simulation experiments, CGFI provides improved finite-sample calibration relative to Wald inference in settings characterized by many estimating equations, non-Gaussian disturbances, dependence among moments, and heteroskedasticity. The method performs particularly well when uncertainty arises from the joint distribution of the estimating equations, which is the setting targeted by the proposed Gaussian copula construction.

The simulations also highlight the limitations of CGFI. Similar to other estimating-equation methods, CGFI does not resolve problems arising from weak identification. In addition, bootstrap inference generally achieves strong coverage, although at substantially higher computational cost. These results suggest that CGFI provides a computationally attractive alternative for regularly identified estimating-equation models where finite-sample uncertainty quantification is important.

6 Conclusion

This paper develops a Copula Generalized Fiducial Inference (CGFI) framework for likelihood-free inference in estimating-equation models. Unlike classical fiducial methods, which require an explicit structural equation for the observed data, the proposed approach constructs a fiducial distribution directly from the empirical estimating equations. By modeling the joint uncertainty of the empirical moments through a Gaussian copula and inverting perturbed estimating equations, CGFI provides a general confidence distribution without requiring either a likelihood function or prior specification.

The theoretical results establish that the proposed procedure is asymptotically valid under standard regularity conditions. Specifically, the CGFI distribution is shown to be consistent and to satisfy a Bernstein–von Mises-type property. The resulting stochastic expansion provides a foundation for future investigation of higher-order refinements. These results demonstrate that the conditional distribution of CGFI draws asymptotically reproduces the sampling distribution of the corresponding estimating-equation estimator while providing a likelihood-free interpretation of parameter uncertainty.

Simulation studies demonstrate that CGFI performs well in finite samples. Across a range of nonlinear estimating-equation models, the proposed method maintains coverage probabilities close to the nominal level even as the number of moment conditions increases substantially. In contrast, conventional Wald confidence intervals exhibit increasing undercoverage as the dimension of the estimating-equation system grows. Bootstrap inference provides comparable statistical performance, but at substantially greater computational cost. The computational advantage of CGFI becomes increasingly pronounced as the dimension of the moment system increases because the method operates directly on the estimated uncertainty of the empirical estimating equations rather than repeatedly resampling and re-estimating the original model.

The proposed framework also opens several directions for future research. One important extension is the development of CGFI for dependent data, including time-series and panel-data estimating equations, where serial dependence must be incorporated into the perturbation mechanism. Another promising direction is the application of CGFI to modern machine-learning estimators that are naturally characterized through estimating equations but for which reliable uncertainty quantification remains computationally challenging. Extending the copula construction beyond the Gaussian case to accommodate heavy-tailed or asymmetric dependence structures also represents an interesting avenue for future work.

More broadly, the proposed framework suggests a different perspective on fiducial inference. Rather than requiring inversion of a structural equation for the observed data, uncertainty can be

constructed from the empirical estimating equations that define the parameter of interest. This substantially enlarges the class of statistical procedures to which fiducial methods can be applied. We hope that this perspective encourages further development of likelihood-free confidence distributions for increasingly complex statistical and econometric models.

Disclosure Statement

The author reports there are no competing interests to declare.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the author used ChatGPT (GPT-5.5-mini, OpenAI) to assist with language editing, organization, and refinement of the manuscript. The author reviewed and revised all AI-generated suggestions and takes full responsibility for the content of the published article.

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